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Reflection

Completing the Bias-Variance Tradeoff lab helped me better understand why choosing the correct level of model complexity is important in machine learning. Before completing this lab, I understood that models could make incorrect predictions, but I did not fully understand how a model could be too simple or too complex. The lab allowed me to see these concepts visually by comparing polynomial regression models with degrees 1, 4, and 15. Using graphs made the differences between underfitting, a good fit, and overfitting much easier for me to recognize.

The degree 1 model was an example of underfitting because it created a straight line that could not follow the curved sine-wave pattern in the data. This model had high bias because it made a simple assumption about the relationship between the variables. It performed poorly because it did not capture enough of the true pattern. The degree 4 model provided the best overall fit because it followed the main shape of the data without trying to match every individual point. This showed me that a useful model does not need to fit every training example perfectly. Instead, it should identify the general pattern so that it can make good predictions with new data.

The degree 15 model demonstrated overfitting. Its prediction line contained many sharp curves because it tried to follow the noise and random changes in the training data. Although this model could perform very well on the data used to train it, it may perform poorly on new data. This represents high variance because the model's results can change greatly when it receives a different training dataset. This part of the lab taught me that a highly complex model is not always the best model.

The learning curves also helped me understand how training and cross-validation scores can be used to identify model problems. For the degree 1 model, the training and cross-validation scores were both relatively low and close together. This confirmed that the model was underfitting and that simply adding more training data would probably not solve the problem. For the degree 15 model, there was a larger gap between the training and cross-validation scores. This showed that the model learned the training data closely but did not generalize as well.

Overall, this lab improved my understanding of the bias-variance tradeoff. I learned that machine learning requires finding a balance between a model that is too simple and one that is too complex. In future projects, I can use validation results and learning curves to select a model that captures important patterns while still performing well on new data.